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LETTERS TO THE EDITOR

AUTHORS' REPLY

HOW MUCH OF THE PLACEBO 'EFFECT' IS REALLY STATISTICAL REGRESSION?

by C. J. McDonald, S. A. Mazzuca and G. P. McCabe, *Statistics in Medicine*, **2**, 417–427 (1983)

We appreciate Dr. Senn's very careful reading of our paper. He picked up an important typographical error in equations (1) and (2).

There are two ways to calculate the effect of statistical regression.

Approach A: select cases at $\mu + k\sigma$
Approach B: select cases at $\mu + k\sigma$ or above.

Equation (1) in the paper uses approach A, and Senn is correct in pointing out the typographical error in our equation (2). In our equations the conditioning expressions should have read $X_1 = X_1$, not $X_1 > X_1$. Then the right side of equation (2) is appropriate as shown in Senn's third equation.

In preparing the manuscript, we calculated the results using both approaches A and B. We presented the equations of approach A because they had a much simpler form. However, we presented the calculations based on approach B – this was an error. However, for $k = 3$ or greater, the two approaches provide very similar results. The equation for percentage improvements using approach A.

$$\text{Equation A: } P_{(a)} = 100\sigma(1-\rho) \Bigg/ \left(\mu\left(\frac{1}{k}\right) + \sigma\right).$$

The equation for approach B is

$$\text{Equation B: } P_{(b)} = 100\,\sigma(1-\rho) \Bigg/ \left(\mu\left[\frac{1-\Phi(k)}{\phi(k)}\right] + \sigma\right).$$

As k gets large, $[1-\Phi(k)]/\phi(k)$ approaches $1/k$.

We used a value of $k = 3$. Substituting 3 into equation A results in 0·3, substituting 3 into equation B gives us 0·33. Moreover, none of these issues changes the conclusions of our paper: statistical regression is large enough to explain the amount of 'improvement' that is often attributed to a placebo effect.

Dr. Senn's other points are also cogent, but we were trying to make the point that techniques exist to reduce the effects of statistical regression. Indeed, under certain assumptions, statistical regression can be eliminated by taking two pretreatment measures, the first to select patients for treatment and the other as the baseline measure, or it can also be reduced by taking a long series of repeat measurements. Moreover, these techniques apply to both the design of clinical trials on many patients and therapeutic trials on individual patients.

We agree with Dr. Senn that in many real circumstances other factors he described would interfere with these techniques. The practical effect of the deviations from theoretics would require information of the components of variance of the magnitude of terms such as $\rho_{12} - \rho_{13}$.

REFERENCES

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CORRECTING FOR THE BIAS CAUSED BY DROPOUTS IN HYPERTENSION TRIALS

by Gordon D. Murray and Janet G. Findlay, *Statistics in Medicine*, 7, 941–946 (1988)

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The recent paper by Murray and Findlay examines a relatively unexplored way of dealing with withdrawals from clinical trials. New insights into this difficult area are very welcome in the pharmaceutical industry because withdrawals pose a common problem when preparing clinical trial results for regulatory submission. The approach they propose has the advantage of accounting for all the patients in the analysis and using all available data and is, in this sense, striving towards the 'intention to treat' ideal. In addition it is optimal under a set of plausible assumptions.

However, I can foresee some difficulties with gaining acceptance of conclusions drawn by means of this methodology. Previous attempts by a colleague and myself[1] to propose related methods were met with considerable scepticism within the pharmaceutical industry. (We suggested that final values of blood pressure in patients withdrawn at an intermediate visit because of uncontrolled blood pressure be estimated by regression from their values at the intermediate visit). This scepticism was connected with the apparent 'creation of data', and the dependence of the method on certain untestable(?) assumptions. The same considerations would seem to apply to Murray and Findlay's suggestion; the assumption of 'missing at random' may be a case in point if the circumstances surrounding withdrawal are not as clear-cut as in their example.

Thus, although I am supportive of their approach, I would not like to rely on its acceptance to achieve successful registration of a new drug. Another valuable proposal for handling withdrawals, which has struggled to establish a place in regulatory work, is that of Gould.[2] For regulatory purposes a simple method commonly known as 'last-value' (LV) analysis is popular in the industry. In this method missing values at the final time of measurement are replaced by the last available value for each patient, generally the one recorded at withdrawal. It would be interesting to know how this method performs on the data in Murray and Findlay's paper. Visual inspection suggests that it might give similar results to the maximum likelihood (ML) approach. If so, it would have the advantage of being conceptually and computationally simpler.